

ESKASONI FIRST NATION, NS, Canada

WMO: 711750

Lat: 45.921N

Lon: 60.647W

Elev: 28

StdP: 100.99

Time Zone: -4.00 (NAA)

Period: 12-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.9	-12.8	-19.5	0.7	-13.4	-17.6	0.8	-11.8	18.9	7.3	16.0	-0.2	3.8	290	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.4	25.7	20.0	24.4	19.4	23.2	18.9	21.6	24.2	20.7	23.0	20.1	22.0	4.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.4	22.8	20.0	14.7	22.0	19.3	14.1	21.5	62.9	24.1	60.1	23.2	57.6	22.0	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	10.4	8.7	DB	-17.7	29.2	2.1	1.3	-19.2	30.1	-20.4	30.9	-21.5	31.6	-23.0	32.5	(4)
(5)			WB	-18.2	22.7	2.0	0.8	-19.6	23.3	-20.8	23.7	-21.9	24.1	-23.3	24.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-3.3	-4.1	-1.7	3.4	8.9	13.3	18.8	19.6	15.5	10.2	4.8	-0.4	(6)	
	DBStd	8.93	4.44	4.66	4.21	3.50	3.08	3.03	2.43	2.14	3.20	3.38	3.97	4.22	(7)	
	HDD10.0	1962	414	394	364	199	59	6	0	0	2	39	163	322	(8)	
	HDD18.3	4182	672	627	622	448	292	153	23	12	94	252	407	580	(9)	
	CDD10.0	922	0	0	1	2	25	104	273	299	167	46	6	0	(10)	
	CDD18.3	101	0	0	0	0	0	2	37	53	9	0	0	0	(11)	
	CDH23.3	249	0	0	0	0	4	8	118	108	11	0	0	0	(12)	
	CDH26.7	22	0	0	0	0	1	1	15	6	0	0	0	0	(13)	
(14) Wind	WSAvg	4.0	4.6	4.4	4.3	4.1	3.6	3.4	3.4	3.6	3.8	3.9	4.4	4.2	(14)	
(15) Precipitation	PrecAvg	1500	148	131	117	112	101	104	88	100	119	153	153	172	(15)	
	PrecMax	1851	227	182	221	187	177	173	170	228	308	267	241	355	(16)	
	PrecMin	1151	70	48	47	59	49	51	37	44	43	61	96	97	(17)	
	PrecStd	152	38	38	42	33	36	37	30	34	60	56	40	53	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	8.7	12.0	14.5	21.7	24.5	28.7	27.5	24.7	19.8	15.9	11.3	(19)	
		MCWB	8.5	8.1	8.7	11.3	15.8	17.6	21.3	21.6	19.8	17.2	14.5	10.3	(20)	
	2%	DB	7.1	6.7	7.0	12.3	18.6	21.7	25.7	25.8	22.7	18.2	14.5	9.7	(21)	
		MCWB	6.8	6.1	5.0	9.4	12.8	16.1	19.9	20.4	18.9	15.8	13.5	9.2	(22)	
	5%	DB	5.2	4.6	5.3	10.6	16.2	20.2	24.6	24.5	21.5	17.0	12.9	8.1	(23)	
		MCWB	4.7	3.8	4.0	8.8	11.3	15.4	19.2	19.8	18.7	14.9	12.0	7.5	(24)	
	10%	DB	3.2	2.7	3.6	8.8	14.2	18.5	23.2	23.4	20.5	15.7	11.0	5.9	(25)	
		MCWB	2.4	2.0	1.9	6.7	10.5	14.5	18.6	19.3	18.2	13.7	9.7	5.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	8.4	9.2	12.6	16.8	19.0	22.7	22.6	21.1	18.0	15.1	10.8	(27)	
		MCDB	8.8	8.6	11.4	13.6	21.0	22.5	26.4	25.2	22.8	18.7	15.7	10.9	(28)	
	2%	WB	6.8	6.3	6.0	10.8	14.3	17.5	21.5	21.8	20.2	16.6	13.7	9.4	(29)	
		MCDB	7.0	6.6	6.5	11.7	16.8	19.8	24.6	24.1	21.6	17.8	14.3	9.7	(30)	
	5%	WB	4.8	4.0	4.1	8.9	12.6	16.5	20.5	21.1	19.7	15.4	12.0	7.5	(31)	
		MCDB	5.1	4.4	5.2	10.0	14.4	18.9	23.0	23.5	21.0	16.6	12.9	8.0	(32)	
	10%	WB	2.6	2.0	2.3	7.4	11.2	15.5	19.8	20.3	18.8	14.1	9.9	5.3	(33)	
		MCDB	3.1	2.6	3.2	8.6	13.2	18.0	22.0	22.5	20.3	15.4	10.8	6.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.8	7.3	7.3	8.6	8.1	7.9	7.4	8.0	7.2	6.4	5.7	(35)	
		MCDBR	9.2	9.9	9.2	9.3	12.1	10.7	9.9	8.5	8.5	7.7	8.7	8.2	(36)	
	5% WB	MCWBR	9.7	9.7	8.1	6.6	7.2	6.3	5.3	4.7	5.8	5.9	8.8	8.7	(37)	
		MCWBR	9.3	10.3	8.9	7.5	9.6	8.9	7.9	7.0	7.2	7.2	8.3	8.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.303	0.295	0.325	0.357	0.370	0.394	0.421	0.397	0.364	0.340	0.323	0.301	(40)	
		taud	2.355	2.408	2.401	2.402	2.425	2.383	2.344	2.411	2.480	2.522	2.502	2.434	(41)	
	Edn at Noon		794	889	905	902	898	874	843	852	854	826	774	761	(42)	
		Edh at Noon	75	86	101	110	112	117	120	109	93	78	66	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.11	1.95	3.20	4.17	5.01	5.38	5.39	4.84	3.78	2.36	1.27	0.83	(44)	
	RadStd		0.08	0.22	0.26	0.34	0.47	0.47	0.36	0.32	0.28	0.21	0.15	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.44	N/A	N/A	N/A	N/A

Nomenclature: See separate page